

SECTION 15170-A MOTOR

1.0 GENERAL

1.1 Description

This Section specifies the requirements for supply, installation, connection and testing of motors.

1.2 Quality Assurance

- A. Provide motors which are the product of a manufacturer who has been engaged in the production of similar motors for a minimum of five years.
- B. Use the appropriate reference codes and specifications of CNS, IEC or NEMA and the codes and regulations of jurisdictional authorities.
- C. Attach name plates securely onto each motor showing manufacturer's name, serial model number and information in accordance with the reference codes.

1.3 Submittals

- A. Shop drawings and manufacturer's literature;
- B. Test reports;
- C. Maintenance data and operating instructions including the following:
 - 1. Description of the motor and its components.
 - 2. Motors operating and maintenance instructions, parts list, illustrations, and diagram for components.
 - 3. Recommended list of spare parts.
 - 4. Wiring diagram.
 - 5. Speed torque curve.
 - 6. Outline and base dimensions.

1.4 Product Delivery, Storage and Handling

- A. Ship each motor securely wrapped, crated or packaged, and labeled for safe handling in shipment and to avoid damage or distortion.
- B. Store motors in a secure and dry storage area.

2.0 PRODUCTS

- 2.1 Electric Motors: Design and performance of motors provided for mechanical equipment shall meet NEMA Standards, shall be energy efficient type, and be suitable for service and location shown or required. Where motors with greasable bearings are called for, they shall be provided with suitable grease fittings.

A. Design Requirements:

1. Motor Voltage Ratings:

Size -KW	Rating Voltage	Phase
below 0.75	220	1 or 3
0.4 to 112	480	3
150 and above	480	3

2. Motor full load power factors:

Size -KW	Power Factor	2 poles	4 poles	6 poles
below 1.5	>80 %	>70 %	>65 %	
1.5 to 5.5	>85 %	>80 %	>70 %	
5.5 to 55	>88 %	>85 %	>80 %	
55 to 200	>88.5 %	>86 %	>85 %	
above 200	>89.5 %	>87 %	>86 %	

3. Provide high efficient motors. The motor efficiency shall be equal or better to NEMA MG 1-1998 Table 12-12. "NEMA Premium".
4. Motor protection: Motors installed in machine room shall be IP 54; motors installed outdoors shall be IP 66 or better protection.
5. Motors for speed-controlled devices shall meet the characteristic performances of the devices and comply with adjustable speed drives.
6. When specified, motors larger than 375kw shall be suitable for closed transition, reduced voltage auto transformer starting.
7. Motors shall have proper characteristics to suit specified makes of equipment. Motors and accessories

shall comply in all respects with ANSI, IEEE, NFPA 70 and NEMA Standards.

8. Over-voltage and over-current of motors shall be protected by MCCs or motor controllers.
9. Motors shall be capable of starting 6 times (constant period) in one hour without overheating.
10.
 - a. Unless otherwise required by duty, location, equipment service, etc., all motors shall be drip-proof, squirrel cage, induction motors, NEMA Design B, having Class F insulation suitable for continuous duty at 40°C ambient and 80°C temperature rise. Over size motors to comply with this requirement are not acceptable. All motors located in the airstream in fan compartment shall be totally enclosed, fan-cooled unless specified otherwise. Where explosion-proof motors are required, they shall be Class I, Division 1, Group D motors.
 - b. Unless otherwise specified, motors for outdoor use shall have NEMA Type II weather-protected enclosures in types and sizes available. Motor sizes not available in NEMA Type II shall be totally enclosed fan-cooled (TEFC). All totally enclosed fan-cooled motors shall have at least one drain with breather plug.
 - c. Surge protection devices shall be insulated from the enclosure and grounded on a visible removable link.
 - d. For the inverter duty application the wire shall be wound with Inverter Spike Resistant magnet wire on designs bellow 2000 volts.
11. Fan motors rated 200 hp and above, shall be NEMA Design B with locked rotor torque in accordance with the latest NEMA Standard MG 1, and with the lock rotor current not to exceed 5 times the full load current. All other motors shall have a minimum locked rotor torque of 100 percent full load torque. Poly-phase induction motors listed in NEMA MG 13 shall be NEMA Design B unless otherwise specified. Single-phase motors shall be split-phase or capacitor start, NEMA Design N unless otherwise specified.
12. Locked-rotor current of three phase integral horsepower motors shall not exceed values listed in NEMA MG 1-12.3, except as specified in Paragraph A.5.

Locked-rotor current of single-phase motors shall not exceed values listed in NEMA MG 1-12.32.

13. Motor service factor shall be 1.15 for open drip-proof motors and 1.15 for totally enclosed motors unless otherwise specified.
14. With 70 percent motor nameplate voltage at the motor terminals, motors shall be capable of accelerating the load driven without exceeding maximum allowable acceleration time.
15. Motor vibration at the rated speed shall not exceed the values given in NEMA MG 1-12.05 or MG 1-20.52.
16. Motors shall operate with an overall sound pressure level that will not require protective equipment. Maximum allowable equipment noise level shall include motor noise.
17. For motors 75kw and larger, not listed in NEMA MG 13, the following performance will be required:
 - a. With the percentage of rated motor voltage specified on the motor nameplate, motors shall accelerate the driven load without exceeding the maximum allowable acceleration time.
 - b. Motors shall be suitable for full voltage across-the-line starting without the starting current exceeding 6 times full load current at rated motor voltage;
 - c. Motors shall have starting duties from ambient temperature with standard wk2 load in accordance with NEMA MG 1-20.43.
18. Bearings and Lubrication
 - a. Unless otherwise specified, motors shall be equipped with antifriction bearings.
 - b. Especially designed larger horsepower motors may include sleeve bearings, if use of such bearings is manufacturer's standard practice.
 - c. Sleeve bearings may have oil-ring or forced-feed lubrication at the manufacturer's option. The forced lubrication system and accessories shall be furnished by the equipment manufacturer.
 - d. Antifriction bearings shall be pre-lubricated, shall be sealed to prevent leakage of lubricant or entrance of foreign matter along the shaft, and shall include provisions for positive in-service

lubrication and a drain to guard against over lubrication.

- e. All motors supplied with antifriction bearings shall have the appropriate AFBP4A (Anti-Friction Bearing Manufacturers Association, Inc.) number stamped on a nameplate attached to the motor. The design of these bearings shall be based on AFBMA standards. Unless otherwise specified, each bearing shall have a minimum L-50 bearing life of 200,000 hours for direct drive.
- f. The rotor of horizontal motors with sleeve bearings shall not ride continuously against either bearing because of thrust developed by the motor. If necessary or required, limited-end-float couplings shall be provided in the driven equipment.

19. Terminal boxes of motors listed in NEMA MG 13 shall be located in accordance with NEMA standards and shall be minimum one size larger than NEMA standard.

20. Terminal boxes of motors must be included terminal base.

21. If the motor has already matched inverter, must use inverter appropriation motor.

B. Rating:

Motor shop drawings shall clearly indicate the NEMA design letter, locked-rotor kVA code letter, frame size, horsepower, speed, voltage, locked-rotor current, full load current, starting torque characteristics, type of windings and insulation, type of bearings and rate life, service factor, nameplate, time rating, and manufacturer's name in English.

1. Voltage and Tolerances:

Motors, starters, and controllers shall meet the specified performance throughout $\pm 10\%$ voltage variation and the characteristics performance of SEMI F 47.

2. Motor Nameplate: The motor nameplate and connection diagram shall be stainless steel, with the following and shall be marked information.

- a. Manufacturer's name
- b. Rated volts and full load current

- c. Rated frequency and number of phases
 - d. Rated full load speed
 - e. Rated temperature rise and rated ambient temperature
 - f. Time rating
 - g. Rated horsepower
 - h. Locked rotor code letter
 - (1) Motors starting on the connection and running on delta, shall be marked with the code letter corresponding to the wye connection.
 - (2) Dual-voltage motors which have a different-locked rotor kVA on the two voltages shall be marked with the code letter for the voltage giving the highest locked-rotor kVA per horsepower (kilowatt).
 - (3) Part-winding-start motors shall be marked with a code letter designating the locked-rotor kVA per horsepower (kilowatt) that is based upon the locked-rotor current for the full winding of the motor.
 - i. NEMA design letter
 - j. Service factor
 - k. on unidirectional motors, an arrow shall be provided to indicate the direction of rotation.
 - 1. A nameplate stating the starting duty limitation shall be mounted on the motor adjacent to the standard motor nameplate.
- 3. In general, motors shall be furnished integrally mounted on all items of mechanical equipment.
 - 4. Motor terminal boxes shall be sized to accommodate the conduit, wire size, and compression type terminal lugs and over sized in accordance with Paragraph 2.1.A.16.
 - 5. Where parallel feeders are required, motor terminal boxes shall be equipped with copper bus bars (size and number as required) to facilitate mounting feeder cable lugs to motor.
 - 6. Painting of motors shall be corrosive resistant with guarantee of 5 years indoor service or 3~5 years outdoor service at least.

C. Accessories

- 1. Integral horsepower motors shall have lifting devices for installing or removing the motor.

2. Slide bases or rails shall be furnished if indicated.
3. Motor frames shall have provisions for frame grounding. For motors with full load current that exceeds 20 amperes, a separate grounding terminal shall be provided within the terminal box.
4. Motors 200 hp and larger not listed in NEMA MG 13 shall be equipped with temperature detectors wired to a separate terminal box unless otherwise shown or specified.
5. Chiller Compressor Motors shall be provided with the following:
 - a. six (6) stator, resistance type, temperature detectors, distributed evenly among the three phases and cooling zones around the stator periphery, copper 10 ohms at 25°C.
 - b. One (1) set of surge protection apparatus, consisting of capacitors and lightning arresters, mounted in a separate, dustproof, indoor type, sheet steel compartment for field mounting adjacent to the motor.
 - c. Three (3) current transformers, 50/5A, for stator differential protection.
 - d. Power Factor capacitors as specified in Paragraph 2.1.C.7.
 - e. One (1) set of space heaters, located in the frame bottom and connected for 110volt, single-phase operation. Heaters shall be capable of maintaining the motor internal temperature at approximately 3 °F above the ambient.
6. All motors 200 HP and larger should be provided with surge packs as specified in Paragraph 2.1.C.7.
7. Power Factor Capacitors

General: Shunt power factor capacitors shall be provided for motors 500 HP and above or specified for improvement of power factor to 0.95.

 - a. Capacitors: Size of capacitors shall be maximum possible in accordance with motor magnetizing due to self excitation and transient torques generated when capacitor and motor are switched as a unit.

The following characteristics shall apply to all the shunt capacitors to be furnished:

- (1) Manufactured and tested in accordance with ANSI Standard C55.1-1968.
- (2) Operating voltage will be as shown on single lines.
- (3) Units shall be integrally fused.
- (4) Capacitors shall discharge to 50 volts or less in 5 minutes, when disconnected from the source of supply.
- (5) Units shall be insulated with nonflammable liquid. Enclosures shall be dust proof suitable for indoor application.
- (6) Lugs shall be provided for wiring as required.

b. Location: Capacitors shall be located adjacent to the motor.

3.0 EXECUTION

- 3.1 Install motors as required by the driven equipment.
- 3.2 Motors shall be given routine tests in accordance with Test Procedures for Poly-phase Induction Motors (NEMA MG 1 and IEEE 112A) or Test Procedures for Single Phase Induction Motors (IEEE 114).
- 3.3 Motor of each size listed in NEMA MG 13 shall be tested in accordance with NEMA standards as follows:
 - A. MG 1-12.42 Temperature Rise;
 - B. MG 1-12.32 (Single Phase) and MG 1-12.36 (Polyphase) Fractional hp Torques;
 - C. MG 1-12.37 to MG 1-12.40 Integral hp Torques;
 - D. MG 1-12.43 Variation from Rated Voltage;
 - E. MG 1-12.44 Variation from Rated Frequency;
 - F. MG 1-12.51 Routing Tests for Polyphase Integral Horsepower Induction Motors.
 - G. Conduct noise test with report (Per NEMA MG 1 Part 9.
- 3.4 Motors 100 hp and larger not listed in NEMA MG 13 shall be tested in accordance with IEEE 112 as follows:
 - A. Routine Test;
 - B. Temperature Rise Test;

- C. Starting Torque Test;
 - D. Efficiency Test at 100-75-50-0 percent loads;
 - E. Power Factor Test at 100-75-50-0 percent loads;
 - F. Resistance Measurement.
 - G. Conduct noise test with report (Per NEMA MG 1 Part 9.)
- 3.5 Motors shall be tested in accordance with IEEE 85, Test Procedure for Airborne Measurements on Rotating Electrical Machinery.

Appendix A:

NEMA MG 1-1998 Table 12-12 (continued) Full-Load Efficiencies for NEMA Premium™ Efficiency Electric Motors Rated 600 Volts or Less (Random Wound)						
ENCLOSED MOTORS						
HP	2 POLE		4 POLE		6 POLE	
	Nominal Efficiency	Minimum Efficiency	Nominal Efficiency	Minimum Efficiency	Nominal Efficiency	Minimum Efficiency
1	77.0	74.0	85.5	82.5	82.5	80.0
1.5	84.0	81.5	86.5	84.0	87.5	85.5
2	85.5	82.5	86.5	84.0	88.5	86.5
3	86.5	84.0	89.5	87.5	89.5	87.5
5	88.5	86.5	89.5	87.5	89.5	87.5
7.5	89.5	87.5	91.7	90.2	91.0	89.5
10	90.2	88.5	91.7	90.2	91.0	89.5
15	91.0	89.5	92.4	91.0	91.7	90.2
20	91.0	89.5	93.0	91.7	91.7	90.2
25	91.7	90.2	93.6	92.4	93.0	91.7
30	91.7	90.2	93.6	92.4	93.0	91.7
40	92.4	91.0	94.1	93.0	94.1	93.0
50	93.0	91.7	94.5	93.6	94.1	93.0
60	93.6	92.4	95.0	94.1	94.5	93.6
75	93.6	92.4	95.4	94.5	94.5	93.6
100	94.1	93.0	95.4	94.5	95.0	94.1
125	95.0	94.1	95.4	94.5	95.0	94.1
150	95.0	94.1	95.8	95.0	95.8	95.0
200	95.4	94.5	96.2	95.4	95.8	95.0
250	95.8	95.0	96.2	95.4	95.8	95.0
300	95.8	95.0	96.2	95.4	95.8	95.0
350	95.8	95.0	96.2	95.4	95.8	95.0
400	95.8	95.0	96.2	95.4	95.8	95.0
450	95.8	95.0	96.2	95.4	95.8	95.0
500	95.8	95.0	96.2	95.4	95.8	95.0

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